Putnam

Probability and expectation

Written by a model for Adit. Not Adit's words.

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1 Probability and expectation

Recognise it. The words random, expected, probability, on average, or almost surely appear. A counting problem asks for a proportion, which is a probability after dividing by the total. A process runs in steps and a stopping time or a final state is asked about. Points are chosen in a region and a geometric condition is imposed. Behind most Putnam probability there is no measure theory at all: a finite sample space, linearity of expectation, a recurrence from the first step, or an area.

Definition 1.1. On a finite or countable sample space Ω with probabilities $p(\omega) \geq 0$ summing to 1, a random variable is a function $X\colon \Omega \to \mathbb{R}$, its expectation is $\mathbb{E}X = \sum_\omega X(\omega)p(\omega)$ when the sum converges absolutely, and the indicator of an event $A \subseteq \Omega$ is $\mathbf{1}_A$, with $\mathbb{E}\,\mathbf{1}_A = \Pr(A)$. The conditional probability of A given B with $\Pr(B) > 0$ is $\Pr(A \mid B) = \Pr(A \cap B)/\Pr(B)$, and $\mathbb{E}(X \mid B)$ is the expectation under the probabilities $p(\omega)/\Pr(B)$ on B.

Theorem 1.2 (Linearity and indicators)**.** $\mathbb{E}(aX + bY) = a\mathbb{E}X + b\mathbb{E}Y$ *for any random variables with finite expectation, independent or not. If* $N = \sum_{i=1}^{n} \mathbf{1}_{A_i}$ *counts how many of the events* A_i *occur, then*

$$\mathbb{E}N = \sum_i \Pr(A_i), \qquad \mathbb{E}N^2 = \sum_i \Pr(A_i) + 2\sum_{i<j} \Pr(A_i \cap A_j).$$

Proof. The first is the distributive law inside the defining sum. For the second, $N^2 = \sum_i \mathbf{1}_{A_i}^2 + 2\sum_{i<j} \mathbf{1}_{A_i}\mathbf{1}_{A_j}$, and $\mathbf{1}_A^2 = \mathbf{1}_A$, $\mathbf{1}_A\mathbf{1}_B = \mathbf{1}_{A\cap B}$. □

The second formula is how a variance is computed in a problem that asks whether a count is concentrated, and it is the only place where the pairwise interactions enter.

Theorem 1.3 (Total probability and conditioning on the first step)**.** *Let* $B_1, B_2, \ldots$ *be a partition of* Ω *into events of positive probability. Then for any event* A *and any random variable* X *with finite expectation,*

$$\Pr(A) = \sum_i \Pr(A \mid B_i)\Pr(B_i), \qquad \mathbb{E}X = \sum_i \mathbb{E}(X \mid B_i)\Pr(B_i).$$

Proof. A is the disjoint union of the $A \cap B_i$, and $\Pr(A \cap B_i) = \Pr(A \mid B_i)\Pr(B_i)$. The expectation version is the same decomposition of the defining sum. □

For a process that moves between finitely many states, condition on the first step: if $h(s)$ is the probability of eventually reaching a target from state s, or the expected number of steps to reach it, then $h(s) = \sum_{s'} P(s \to s')h(s')$, plus 1 in the expected-time case, with h known on the target. This is a linear system, and a Putnam problem is built so that it has a short solution: a symmetry that makes most unknowns equal, or a telescoping along the states.

Proposition 1.4 (Tail sum)**.** *For a random variable N taking values in $\{0, 1, 2, \dots\}$, $\mathbb{E}N = \sum_{k \geq 1} \Pr(N \geq k)$. For $X \geq 0$ with a density, $\mathbb{E}X = \int_0^\infty \Pr(X > t)\, dt$.*

Proof. $N = \sum_{k \geq 1} \mathbf{1}[N \geq k]$ pointwise; take expectations and swap the sum and the expectation, which Tonelli allows since everything is non-negative. The continuous case is the same with an integral. □

Proposition 1.5 (Geometric probability)**.** *If a point is chosen uniformly in a region $R \subset \mathbb{R}^d$ of finite positive volume, the probability that it lies in a measurable $S \subseteq R$ is $\mathrm{vol}(S)/\mathrm{vol}(R)$. If $X_1, \dots, X_d$ are independent and each uniform on $[0, 1]$, then $(X_1, \dots, X_d)$ is uniform on the cube $[0, 1]^d$, so a probability about them is a volume.*

Proof. The first is the definition of the uniform distribution. For the second, independence means the joint density is the product of the marginal densities, which is 1 on the cube. □

Proposition 1.6 (The probabilistic method)**.** *If $\mathbb{E}X = \mu$ then $\Pr(X \geq \mu) > 0$ and $\Pr(X \leq \mu) > 0$. In particular, if a random object has expected score μ, some object has score at least μ. If $X \geq 0$ then $\Pr(X \geq a) \leq \mathbb{E}X/a$ for $a > 0$ (Markov), so if $\mathbb{E}X < 1$ for a non-negative integer X then $\Pr(X = 0) > 0$.*

Proof. If $X < \mu$ everywhere then $\mathbb{E}X < \mu$. For Markov, $a\,\mathbf{1}[X \geq a] \leq X$ pointwise; take expectations. □

First moves. Write the quantity as a sum of indicators and compute one probability. Condition on the first step and write the recurrence. List every outcome of the smallest case and count. For points in a region, draw the region where the condition holds and compute its area; look for a line or a curve as the boundary. For an expected stopping time, ask whether every step has the same chance of stopping, which makes it geometric. For a question of the form "show that some arrangement has property P", choose the arrangement at random and compute the expected number of failures.

Example 1.7. A fair six-sided die is rolled repeatedly and the running total of the rolls is kept. Find the expected number of rolls until the running total is first divisible by 4.

Solution. Only the residue of the total modulo 4 matters. A roll adds a residue $d \in \{0, 1, 2, 3\}$ with probabilities $p_0 = \frac{1}{6}$, $p_1 = \frac{2}{6}$, $p_2 = \frac{2}{6}$, $p_3 = \frac{1}{6}$, since the faces $1, \dots, 6$ reduce to $1, 2, 3, 0, 1, 2$. Let T_r be the expected number of further rolls needed to reach residue 0 from residue r, with $T_0 = 0$, and let $P(r, s) = p_{s-r}$ (indices modulo 4). A step of $+1$

has positive probability, so every residue is reachable from every other and all T_r are finite. Conditioning on the next roll, for $r \neq 0$,

$$T_r = 1 + \sum_s P(r,s)\, T_s.$$

The answer asked for is the expected return time to 0 starting from a fresh total of 0: $T^* = 1 + \sum_s P(0,s) T_s$.

Rather than solving the system, add the three equations for $r = 1, 2, 3$:

$$\sum_{r \neq 0} T_r = 3 + \sum_s T_s \sum_{r \neq 0} P(r,s) = 3 + \sum_s T_s \big(1 - P(0,s)\big),$$

because each column of P sums to 1: $\sum_r P(r,s) = \sum_r p_{s-r} = 1$. Since $T_0 = 0$, the left side is $\sum_s T_s$, and cancelling it gives $\sum_s P(0,s) T_s = 3$. Hence $T^* = 1 + 3 = 4$.

As a check, the system solves to $T_1 = \frac{98}{25}$, $T_2 = \frac{84}{25}$, $T_3 = \frac{86}{25}$, and $1 + \frac{1}{6}\big(2 \cdot \frac{98}{25} + 2 \cdot \frac{84}{25} + \frac{86}{25}\big) = 1 + \frac{450}{150} = 4$. The argument used nothing about the die except that each column of P sums to 1, which holds for any step distribution on any modulus m with the walk irreducible, and gives m every time. That is the uniformity of the previous section hiding in a probability problem: the stationary distribution of a random walk on a group is uniform, so the mean return time to any state is the number of states. □

Example 1.8. Two numbers x and y are chosen independently and uniformly from $[0, 1]$. Find the probability that $x + y > 1$ and $xy < \frac{1}{4}$.

Solution. The pair (x, y) is uniform on the unit square, so the probability is the area of the region $\{x + y > 1,\ xy < \frac{1}{4}\}$. The region $\{x + y > 1\}$ is the triangle above the diagonal, of area $\frac{1}{2}$. Inside it, remove the part where $xy \geq \frac{1}{4}$.

By the arithmetic-geometric mean inequality, $xy \geq \frac{1}{4}$ forces $x + y \geq 2\sqrt{xy} \geq 1$, with equality only at the single point $(\frac{1}{2}, \frac{1}{2})$. So $\{xy \geq \frac{1}{4}\}$ lies inside $\{x + y \geq 1\}$ up to a set of area zero, and the area to remove is the whole of $\{xy \geq \frac{1}{4}\}$ in the square. That region is $\frac{1}{4} \leq x \leq 1$, $\frac{1}{4x} \leq y \leq 1$, with area

$$\int_{1/4}^{1} \Big(1 - \frac{1}{4x}\Big) dx = \frac{3}{4} - \frac{1}{4} \ln 4 = \frac{3}{4} - \frac{\ln 2}{2}.$$

Therefore the probability is

$$\frac{1}{2} - \Big(\frac{3}{4} - \frac{\ln 2}{2}\Big) = \frac{\ln 2}{2} - \frac{1}{4} = \frac{2 \ln 2 - 1}{4} \approx 0.097.$$

It is positive, as it should be since $(0.9, 0.2)$ is in the region, and small, since the hyperbola $xy = \frac{1}{4}$ hugs the line $x + y = 1$ near the centre. The inequality step is what made the picture simple: without it, the region to remove would have had to be intersected with the triangle by hand. □

Trap. Multiplying probabilities of events that share randomness; two positions of a random permutation being fixed are not independent, and the indicator method is exactly the way to avoid needing them to be. A sample space that is not uniform: two dice summing to 7 is six outcomes out of 36, not one of eleven. Conditioning on the first step and forgetting the $+1$ for the step itself. A stopping time whose expectation is infinite, so the recurrence has no finite solution; check irreducibility or a geometric bound first. In geometric probability, a region drawn with a straight boundary where the condition gives a curve. Computing an expectation and reporting it as the most likely value; the expected number of fixed points of a permutation is 1, which says nothing about what happens most often.

Ladder.

1. PS 6 #3. A biased coin tossed many times; the parity of the number of heads.
2. PS 6 #5. Uniform draws added until a threshold is crossed; the expected count.
3. PS 6 #13. A random point in a disc; nearer the centre or the edge.
4. 2021 B1. A random square placed on a grid; the chance it avoids certain points.
5. 2020 A4. Random blackening of a strip; the limiting white fraction.
6. 2020 B3. A randomly shrinking interval; the expected number of steps below a threshold.
7. 2023 B3. A random sequence and its longest alternating subsequence; the expectation.
8. 2024 B4. A walk pulled toward uniform draws; the limiting mean.
9. 2022 A4. A weighted sum of random terms until an ascent; the expectation.
10. 2016 B4. A random antisymmetric matrix; the expected determinant.

Where it hides. Inside counting, by the probabilistic method: a random object has the property with positive probability, so one exists (2024 A3, 2018 B6). Inside recurrences, where the first-step equation is the recurrence (2020 A4, 2020 B3). Inside geometry, where a chance is a ratio of areas (2021 B1, 2024 A5). Inside inequalities, where a bound on an expectation or a tail is the claim (2021 B6, 2022 B5). Inside linear algebra, where the expected determinant of a random matrix is a signed sum of expected products (2016 B4, PS 7 #12). Inside games, where the two players' chances are compared (2017 A5, PS 6 #24 to #28).